Amaan Pathan

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Summary

Systems Engineer with open source experience at BookBrainz and Guidefox and production freelance work at NIAT serving 500+ active users. Proficient in Golang, TypeScript/Node.js, PostgreSQL, Redis, and backend architecture.

Skills

- **Languages:** Golang, TypeScript, JavaScript, Python, SQL
- **Backend:** RESTful APIs, Distributed Systems, RBAC, RabbitMQ, Microservices, Node.js, Express.js, Gin
- **Databases:** MySQL, PostgreSQL, Redis, MongoDB; Schema Design, Query Optimization
- **Infrastructure & DevOps:** Docker, Linux/Unix, Bash, Nginx, Kubernetes, AWS, Cloudflare, CI/CD, Git

Work Experience

Open Source Contributor Jan. 2026 – Mar. 2026
BookBrainz *Remote*

- Replaced external Docker wait-scripts with Docker-native health checks across 3 services (PostgreSQL, Elasticsearch, Redis), eliminating startup race conditions and reducing environment setup failures by ∼90%.
- Resolved GitHub Actions CI failures caused by Docker version incompatibilities on ubuntu-latest runners, restoring a 100% CI pass rate and unblocking the project's active PR review pipeline.

Software Engineer Sep. 2025 – Oct. 2025
NIAT (Freelance / Contract) *Remote*

- Designed RESTful backend services for an internal platform (500+ users): authentication, RBAC, and a RAG pipeline improving query accuracy by 35%.
- Optimized API response times to under 120ms under concurrent load through query tuning and targeted caching strategies.
- Implemented hybrid dense and sparse retrieval with a reranking stage in the RAG pipeline (LlamaIndex, Qdrant, FastAPI), extending document ingestion to PDFs, Word docs, and slide decks alongside voice queries via Whisper transcription

Open Source Contributor Oct. 2024 – Jan. 2025
Guidefox *Remote*

- Integrated Zod-based validation across 20+ APIs, reducing runtime errors by 40%; introduced DTO-based contracts, cutting duplicated logic by 30%.
- Expanded test coverage from 55% to 80% through targeted unit and integration tests, cutting defect escape rate and accelerating CI merge cycles.
- Centralized error handling with a reusable ErrorHandler class and added validation middleware across banner API routes; extended matching error-state handling to the frontend banner forms for consistent user feedback.

Projects

Hermes – Fault-Tolerant Workflow Automation Engine GitHub
Golang · PostgreSQL · Redis · Docker · NATS JetStream

- Built a self-hostable event-driven workflow automation engine with parallel workers handling 1,000+ jobs/hour across multi-step automation pipelines.
- Enforced retries, idempotency, and failure isolation; used NATS JetStream to guarantee at-least-once delivery semantics.

Surang – Self-Hostable Reverse Tunnel Server GitHub | Live
Golang · CI/CD · SQLite · Docker · Docker

- Built a self-hostable ngrok alternative in Go; client dials out over a persistent yamux-multiplexed connection to bypass NAT, with random three-word subdomain routing and a custom dial-frame wire protocol.
- Shipped token-based auth and a TUI client; released cross-platform static binaries via GitHub Actions and a Docker image.

Wingman – AeroMentor Platform Backend GitHub
Golang · PostgreSQL · NATS JetStream · Redis · MinIO · Qdrant · Docker

- Architected the Go backend for AeroMentor (AI-powered LMS) with Clean Architecture; shipped course, lesson, analytics, and JWT/RBAC endpoints with Redis token-bucket rate limiting at 60 req/min per user.
- Built an async RAG ingestion pipeline via NATS JetStream and Qdrant for AI chat and quiz generation; handled file delivery through presigned MinIO URLs and TUS resumable uploads.

Education

B.Tech in Computer Science Engineering 2022 – 2026
University of Lucknow, Lucknow, India